

In the Name of God

Computational Fluid Dynamics II

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Chapter 6

Incompressible Navier-Stokes Equations

فصل ۶

معادلات ناویر-استوکس تراکم ناپذیر

معادلات ناویر-استوکس تراکم‌ناپذیر

- نکات مربوط به معادلات ناویر-استوکس تراکم‌ناپذیر

- نیاز به کوپل کردن معادلات پیوستگی و مومنتوم به روشهای خاص ریاضیاتی
- مستقل بودن معادله انرژی از معادلات پیوستگی و مومنتوم (به استثناء حالت جابجایی آزاد)
- دارای ماهیت بیضوی-سهموی
- حجم و حافظه محاسباتی کمتر نسبت به معادلات ناویر-استوکس تراکم‌پذیر

$$\vec{\nabla} \cdot \vec{V} = 0$$

$$\rho \left(\frac{\partial \vec{V}}{\partial t} + (\vec{V} \cdot \vec{\nabla}) \vec{V} \right) = -\vec{\nabla} p + \mu \nabla^2 \vec{V} + \rho \vec{f}$$

$$\rho c_p \left(\frac{\partial T}{\partial t} + \vec{V} \cdot \vec{\nabla} T \right) = k \nabla^2 T + \Phi$$

معادلات ناویر-استوکس تراکم‌ناپذیر

- معادلات ناویر-استوکس تراکم‌ناپذیر دوبعدی

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

$$\rho \left(\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \right) = -\frac{\partial p}{\partial x} + \mu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) + \rho f_x$$

$$\rho \left(\frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} \right) = -\frac{\partial p}{\partial y} + \mu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right) + \rho f_y$$

$$\rho c_p \left(\frac{\partial T}{\partial t} + u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} \right) = k \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right) + \Phi \quad \& \quad \Phi = \mu \left\{ 2 \left(\frac{\partial u}{\partial x} \right)^2 + 2 \left(\frac{\partial v}{\partial y} \right)^2 + \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right)^2 \right\}$$

معادلات ناویر-استوکس تراکم‌ناپذیر

- بی‌بعدسازی معادلات ناویر-استوکس تراکم‌ناپذیر دوبعدی

$$\tilde{t} = \frac{t}{L/u_\infty}, \quad \tilde{x} = \frac{x}{L}, \quad \tilde{y} = \frac{y}{L}, \quad \tilde{u} = \frac{u}{u_\infty}, \quad \tilde{v} = \frac{v}{u_\infty}, \quad \tilde{p} = \frac{p}{\rho_\infty u_\infty^2}, \quad \tilde{T} = \frac{T}{T_\infty}, \quad \tilde{f} = \frac{f}{g}$$

$$\frac{\partial \tilde{u}}{\partial \tilde{x}} + \frac{\partial \tilde{v}}{\partial \tilde{y}} = 0$$

$$\frac{\partial \tilde{u}}{\partial \tilde{t}} + \tilde{u} \frac{\partial \tilde{u}}{\partial \tilde{x}} + \tilde{v} \frac{\partial \tilde{u}}{\partial \tilde{y}} = -\frac{\partial \tilde{p}}{\partial \tilde{x}} + \frac{1}{\text{Re}} \left(\frac{\partial^2 \tilde{u}}{\partial \tilde{x}^2} + \frac{\partial^2 \tilde{u}}{\partial \tilde{y}^2} \right) + \frac{1}{\text{Fr}} \tilde{f}_x$$

$$\frac{\partial \tilde{v}}{\partial \tilde{t}} + \tilde{u} \frac{\partial \tilde{v}}{\partial \tilde{x}} + \tilde{v} \frac{\partial \tilde{v}}{\partial \tilde{y}} = -\frac{\partial \tilde{p}}{\partial \tilde{y}} + \frac{1}{\text{Re}} \left(\frac{\partial^2 \tilde{v}}{\partial \tilde{x}^2} + \frac{\partial^2 \tilde{v}}{\partial \tilde{y}^2} \right) + \frac{1}{\text{Fr}} \tilde{f}_y$$

$$\frac{\partial \tilde{T}}{\partial \tilde{t}} + \tilde{u} \frac{\partial \tilde{T}}{\partial \tilde{x}} + \tilde{v} \frac{\partial \tilde{T}}{\partial \tilde{y}} = \frac{1}{\text{Re Pr}} \left(\frac{\partial^2 \tilde{T}}{\partial \tilde{x}^2} + \frac{\partial^2 \tilde{T}}{\partial \tilde{y}^2} \right) + \tilde{\Phi} \quad \& \quad \tilde{\Phi} = \frac{\text{Ec}}{\text{Re}} \left\{ 2 \left(\frac{\partial \tilde{u}}{\partial \tilde{x}} \right)^2 + 2 \left(\frac{\partial \tilde{v}}{\partial \tilde{y}} \right)^2 + \left(\frac{\partial \tilde{v}}{\partial \tilde{x}} + \frac{\partial \tilde{u}}{\partial \tilde{y}} \right)^2 \right\}$$

$$\text{Re} = \frac{\rho u_\infty L}{\mu}, \quad \text{Fr} = \frac{u_\infty^2}{gL}, \quad \text{Ec} = \frac{u_\infty^2}{c_p T_\infty}$$

معادلات ناویر-استوکس تراکم‌ناپذیر

• معادلات ناویر-استوکس تراکم‌ناپذیر دوبعدی در فرم بی‌بعد

- حذف $\sim$ از سیستم معادلات

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = -\frac{\partial p}{\partial x} + \frac{1}{\text{Re}} \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) + \frac{1}{\text{Fr}} f_x$$

$$\frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} = -\frac{\partial p}{\partial y} + \frac{1}{\text{Re}} \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right) + \frac{1}{\text{Fr}} f_y$$

$$\frac{\partial T}{\partial t} + u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} = \frac{1}{\text{Re Pr}} \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right) + \tilde{\Phi} \quad \& \quad \tilde{\Phi} = \frac{\text{Ec}}{\text{Re}} \left\{ 2 \left(\frac{\partial u}{\partial x} \right)^2 + 2 \left(\frac{\partial v}{\partial y} \right)^2 + \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right)^2 \right\}$$

معادلات ناویر-استوکس تراکم ناپذیر

• فرمولاسیون چرخش-تابع جریان (Vorticity-Stream Function Formulation)

- معادله حاکم بر تابع جریان

$$u = \frac{\partial \psi}{\partial y} \quad \& \quad v = -\frac{\partial \psi}{\partial x}$$

$$\vec{\zeta} = \vec{\nabla} \times \vec{V} \quad \rightarrow \quad \zeta_z = \zeta = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = -\left(\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} \right) = -\nabla^2 \psi \quad \text{Poisson eq. (elliptic)}$$

- معادله حاکم بر چرخش

$$\frac{\partial}{\partial y}(x\text{-mom.}) - \frac{\partial}{\partial x}(y\text{-mom.}) \quad \rightarrow$$

$$\frac{\partial}{\partial t} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) + u \frac{\partial}{\partial x} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) + v \frac{\partial}{\partial y} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) = \frac{1}{\text{Re}} \left(\frac{\partial^2}{\partial x^2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) + \frac{\partial^2}{\partial y^2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \right)$$

$$\frac{\partial \zeta}{\partial t} + u \frac{\partial \zeta}{\partial x} + v \frac{\partial \zeta}{\partial y} = \frac{1}{\text{Re}} \left(\frac{\partial^2 \zeta}{\partial x^2} + \frac{\partial^2 \zeta}{\partial y^2} \right) \quad \text{vorticity eq. (parabolic)}$$

$$\frac{\partial \zeta}{\partial t} + \frac{\partial(u\zeta)}{\partial x} + \frac{\partial(v\zeta)}{\partial y} = \frac{1}{\text{Re}} \left(\frac{\partial^2 \zeta}{\partial x^2} + \frac{\partial^2 \zeta}{\partial y^2} \right) \quad \text{conservative form}$$

معادلات ناویر-استوکس تراکم‌ناپذیر

- فرمولاسیون چرخش-تابع جریان (Vorticity-Stream Function Formulation)
- مزایای روش
 - استفاده از روشهای متداول جهت حل معادله پواسون تابع جریان و حل معادله سهموی چرخش
 - عدم وجود عبارت فشار در فرمولاسیون روش چرخش-تابع جریان
- معایب روش
 - پیچیدگی زیاد این روش برای تحلیل مسائل سه‌بعدی
 - نیاز به حل معادله پواسون برای فشار
 - عدم وجود شرایط مرزی مستقیم برای چرخش روی مرزها

معادلات ناویر-استوکس تراکم‌ناپذیر

- حل عددی روش چرخش-تابع جریان

- حل معادله تابع جریان به روشهای تکراری

$$\nabla^2 \psi = \frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = -\zeta$$

$$\frac{\psi_{i-1,j} - 2\psi_{i,j} + \psi_{i+1,j}}{(\Delta x)^2} + \frac{\psi_{i,j-1} - 2\psi_{i,j} + \psi_{i,j+1}}{(\Delta y)^2} = -\zeta_{i,j}$$

- حل معادله تابع جریان با انتگرال‌گیری در زمان مجازی

$$\frac{\partial \psi}{\partial \tau} - \left(\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \zeta \right) = 0$$

$$\frac{\psi_{i,j}^{n+1} - \psi_{i,j}^n}{\Delta \tau} - \left(\frac{\psi_{i-1,j}^n - 2\psi_{i,j}^n + \psi_{i+1,j}^n}{(\Delta x)^2} + \frac{\psi_{i,j-1}^n - 2\psi_{i,j}^n + \psi_{i,j+1}^n}{(\Delta y)^2} + \zeta_{i,j}^n \right) = 0$$

معادلات ناویر-استوکس تراکم‌ناپذیر

• حل عددی روش چرخش-تابع جریان

- حل معادله چرخش با انتگرال‌گیری در زمان

$$\frac{\partial \zeta}{\partial t} + u \frac{\partial \zeta}{\partial x} + v \frac{\partial \zeta}{\partial y} = \frac{1}{\text{Re}} \left(\frac{\partial^2 \zeta}{\partial x^2} + \frac{\partial^2 \zeta}{\partial y^2} \right)$$

$$\frac{\zeta_{i,j}^{n+1} - \zeta_{i,j}^n}{\Delta t} + \left(u \frac{\partial \zeta}{\partial x} \right)_{i,j}^n + \left(v \frac{\partial \zeta}{\partial y} \right)_{i,j}^n = \frac{1}{\text{Re}} \left(\frac{\zeta_{i-1,j}^n - 2\zeta_{i,j}^n + \zeta_{i+1,j}^n}{(\Delta x)^2} + \frac{\zeta_{i,j-1}^n - 2\zeta_{i,j}^n + \zeta_{i,j+1}^n}{(\Delta y)^2} \right)$$

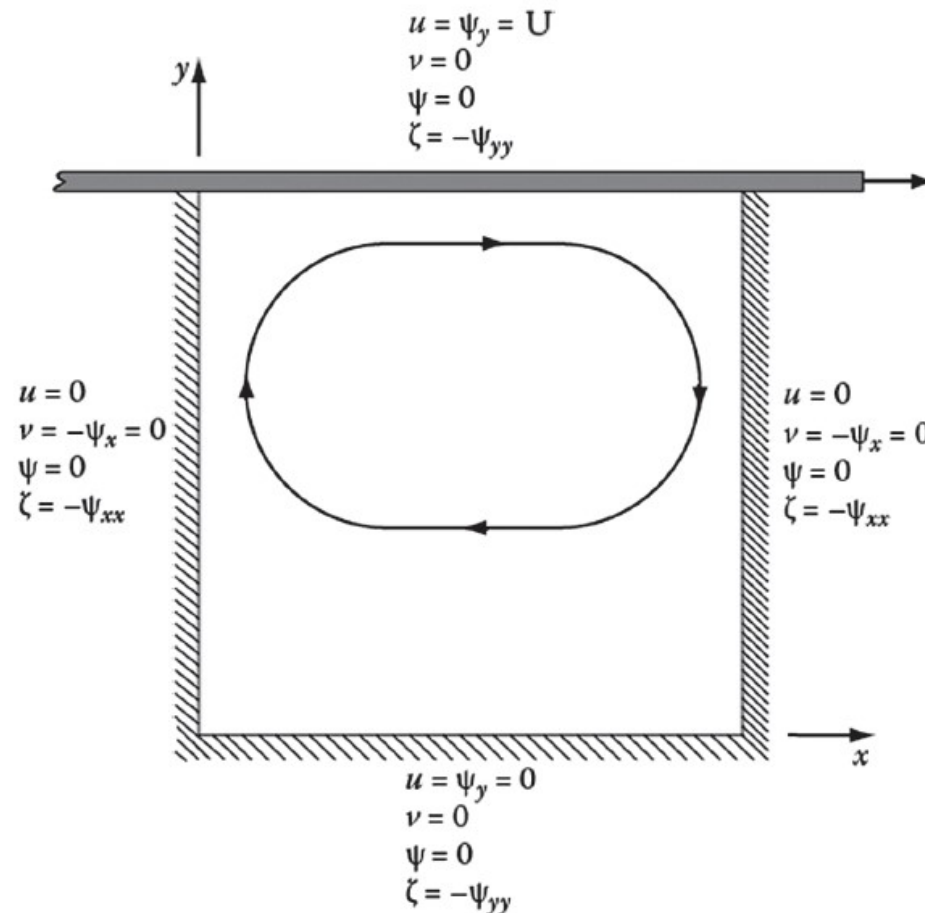
$$\left(u \frac{\partial \zeta}{\partial x} \right)_{i,j}^n = \begin{cases} u_{i,j}^n \frac{\zeta_{i+1,j}^n - \zeta_{i,j}^n}{\Delta x}, & u_{i,j}^n < 0 \\ u_{i,j}^n \frac{\zeta_{i,j}^n - \zeta_{i-1,j}^n}{\Delta x}, & u_{i,j}^n > 0 \end{cases}$$

$$\left(v \frac{\partial \zeta}{\partial y} \right)_{i,j}^n = \begin{cases} v_{i,j}^n \frac{\zeta_{i,j+1}^n - \zeta_{i,j}^n}{\Delta y}, & v_{i,j}^n < 0 \\ v_{i,j}^n \frac{\zeta_{i,j}^n - \zeta_{i,j-1}^n}{\Delta y}, & v_{i,j}^n > 0 \end{cases}$$

معادلات ناویر-استوکس تراکم‌ناپذیر

• فرمولاسیون چرخش-تابع جریان (Vorticity-Stream Function Formulation)

- شرایط مرزی برای مساله حفره سربسته



معادلات ناویر-استوکس تراکم‌ناپذیر

• فرمولاسیون چرخش-تابع جریان (Vorticity-Stream Function Formulation)

- شرایط مرزی برای مساله حفره سر بسته

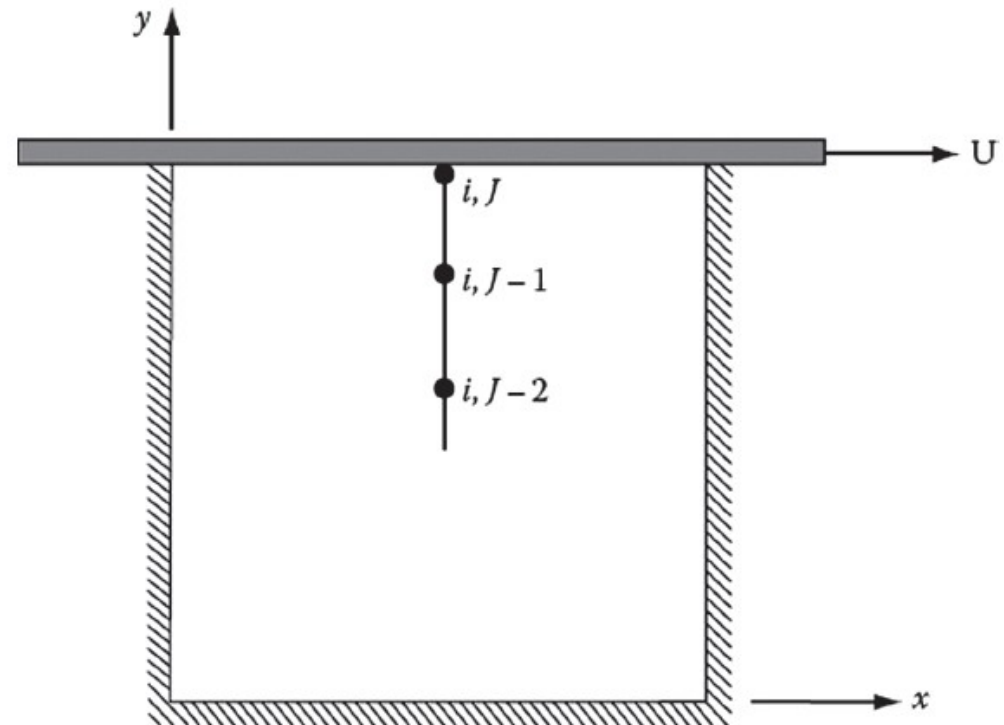
At $y = y_{\max}$

$$\psi_{i,J-1} = \psi_{i,J} - \Delta y \left(\frac{\partial \psi}{\partial y} \right)_{i,J} + \frac{1}{2} (\Delta y)^2 \left(\frac{\partial^2 \psi}{\partial y^2} \right)_{i,J} + \dots$$

$$\left\{ \begin{array}{l} \left(\frac{\partial \psi}{\partial y} \right)_{i,J} = u_{i,J} = U \\ \left(\psi_{i,J} = \frac{\partial^2 \psi}{\partial x^2} \right)_{i,J} = 0 \end{array} \right. \rightarrow \zeta_{i,J} = - \left(\frac{\partial^2 \psi}{\partial y^2} \right)_{i,J}$$

$$\psi_{i,J-1} = 0 - \Delta y U - \frac{1}{2} (\Delta y)^2 \zeta_{i,J} \rightarrow$$

$$\zeta_{i,J} = - \frac{2(\psi_{i,J-1} + \Delta y U)}{(\Delta y)^2}$$



معادلات ناویر-استوکس تراکم‌ناپذیر

- فرمولاسیون چرخش-تابع جریان (Vorticity-Stream Function Formulation)

- معادله فشار (فرم بابت)

$$\frac{\partial}{\partial y}(x - mom.) + \frac{\partial}{\partial x}(y - mom.) \rightarrow$$

$$\nabla^2 p = \frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 p}{\partial y^2} = 2\rho \left(\frac{\partial u}{\partial x} \frac{\partial v}{\partial y} - \frac{\partial v}{\partial x} \frac{\partial u}{\partial y} \right) = 2\rho \left(\frac{\partial^2 \psi}{\partial x^2} \frac{\partial^2 \psi}{\partial y^2} - \frac{\partial^2 \psi}{\partial x \partial y} \right) = S \quad \text{Poisson eq. (elliptic)}$$

- معادله فشار (فرم بی‌بعد)

$$\nabla^2 p = 2 \left(\frac{\partial^2 \psi}{\partial x^2} \frac{\partial^2 \psi}{\partial y^2} - \frac{\partial^2 \psi}{\partial x \partial y} \right) = S$$

- ماهیت بیضوی برای فشار در معادلات جریان تراکم‌ناپذیر

- حل معادله فشار فقط یک بار و بعد از حل معادلات چرخش-تابع جریان

- برای تعیین میدان فشار، مقدار فشار در یک نقطه لازم است معلوم باشد.

معادلات ناویر-استوکس تراکم‌ناپذیر

• فرمولاسیون چرخش-تابع جریان (Vorticity-Stream Function Formulation)

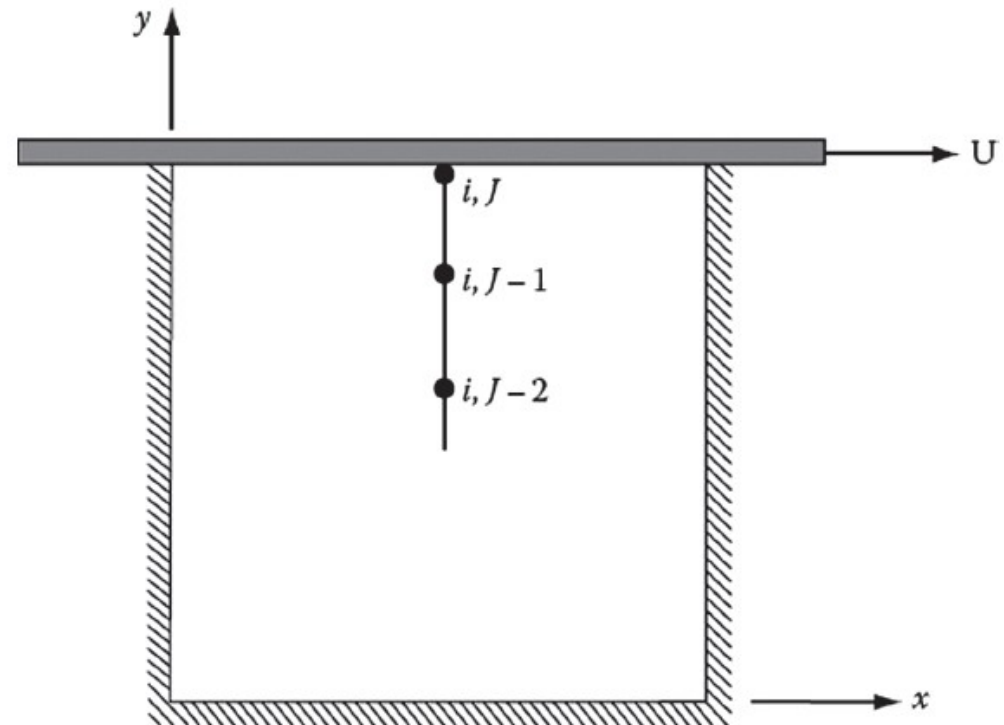
- شرایط مرزی برای مساله حفره سربسته

At $y = y_{\max}$

$$y\text{-mom.: } \left. \frac{\partial p}{\partial y} \right)_{i,J} = \frac{1}{\text{Re}} \left. \frac{\partial^2 v}{\partial y^2} \right)_{i,J}$$

At $x = x_{\min}$

$$x\text{-mom.: } \left. \frac{\partial p}{\partial x} \right)_{1,j} = \frac{1}{\text{Re}} \left. \frac{\partial^2 u}{\partial x^2} \right)_{1,j}$$



معادلات ناویر-استوکس تراکم‌ناپذیر

- روش تراکم‌پذیری مصنوعی (Artificial Compressibility Method)

- اصلاح معادله پیوستگی با اضافه کردن عبارت تراکم‌پذیری مصنوعی

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \quad \rightarrow \quad \frac{\partial \tilde{\rho}}{\partial \tau} + \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \quad \& \quad \tilde{\rho} = \text{artificial density}$$

$$p = \beta \tilde{\rho} = \tilde{a}^2 \tilde{\rho} \quad \text{equation of state (EOS)} \quad \& \quad \begin{cases} \beta = \text{artificial compressibility,} \\ \tilde{a} = \text{artificial sound speed} = \beta^{\frac{1}{2}} \end{cases}$$

$$\tilde{Mach}\big|_{\max} = \tilde{M}\big|_{\max} = \frac{V_{\max}}{\tilde{a}} = \frac{V_{\max}}{\beta^{\frac{1}{2}}} \leq 1 \quad \& \quad V_{\max} = \left(u^2 + v^2\right)^{\frac{1}{2}}$$

- محدوده مناسب تغییرات ضریب تراکم‌پذیری مصنوعی

$$0.1 \leq \beta \leq 10$$

معادلات ناویر-استوکس تراکم ناپذیر

• روش تراکم پذیری مصنوعی (Artificial Compressibility Method)

- سیستم معادلات حاکم

$$\frac{1}{\beta} \frac{\partial p}{\partial \tau} + \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = -\frac{\partial p}{\partial x} + \frac{1}{\text{Re}} \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right)$$

$$\frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} = -\frac{\partial p}{\partial y} + \frac{1}{\text{Re}} \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right)$$

- مقادیر ویژه سیستم معادلات

$$\begin{cases} \lambda|_A = u, u + a_e, u - a_e & \& a_e = (\beta + u^2)^{\frac{1}{2}} \\ \lambda|_B = v, v + a_e, v - a_e & \& a_e = (\beta + v^2)^{\frac{1}{2}} \end{cases} \quad \& \quad a_e = \text{effective acoustic wave speed}$$

- ماهیت سیستم معادلات: بیضوی-سهموی

معادلات ناویر-استوکس تراکم‌ناپذیر

• روش تراکم‌پذیری مصنوعی (Artificial Compressibility Method)

- فرم بقایی سیستم معادلات

$$\frac{\partial p}{\partial \tau} + \beta \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) = 0$$

$$\frac{\partial u}{\partial t} + \frac{\partial (u^2 + p)}{\partial x} + \frac{\partial (uv)}{\partial y} = \frac{1}{\text{Re}} \nabla^2 u$$

$$\frac{\partial v}{\partial t} + \frac{\partial (uv)}{\partial x} + \frac{\partial (v^2 + p)}{\partial y} = \frac{1}{\text{Re}} \nabla^2 v$$

- فرم ماتریسی سیستم معادلات

$$\frac{\partial Q}{\partial t} + \frac{\partial F}{\partial x} + \frac{\partial G}{\partial y} = \frac{1}{\text{Re}} N \nabla^2 Q$$

$$Q = \begin{pmatrix} p \\ u \\ v \end{pmatrix}, \quad F = \begin{pmatrix} \beta u \\ u^2 + p \\ uv \end{pmatrix}, \quad G = \begin{pmatrix} \beta v \\ uv \\ v^2 + p \end{pmatrix}, \quad N = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

معادلات ناویر-استوکس تراکم‌ناپذیر

• روش تراکم‌پذیری مصنوعی (Artificial Compressibility Method)

- روش ضمنی اویلر

$$\frac{\partial Q^{n+1}}{\partial t} + \frac{\partial F^{n+1}}{\partial x} + \frac{\partial G^{n+1}}{\partial y} = \frac{1}{\text{Re}} N \nabla^2 Q^{n+1}$$

$$\begin{aligned} F^{n+1} &= F^n + \left(\frac{\partial F}{\partial t} \right)^n \Delta t + O(\Delta t)^2 = F^n + \left(\frac{\partial F}{\partial Q} \right)^n \frac{\partial Q}{\partial t} \Delta t + O(\Delta t)^2 \\ &= F^n + A^n \left(\frac{Q^{n+1} - Q^n}{\Delta t} + O(\Delta t) \right) \Delta t + O(\Delta t)^2 \end{aligned}$$

$$\begin{cases} F^{n+1} = F^n + A^n \Delta Q^n + O(\Delta t)^2 \\ G^{n+1} = G^n + B^n \Delta Q^n + O(\Delta t)^2 \end{cases} \quad \& \quad \Delta Q^n = Q^{n+1} - Q^n$$

$$A = \frac{\partial F}{\partial Q} = \begin{pmatrix} 0 & \beta & 0 \\ 1 & 2u & 0 \\ 0 & v & u \end{pmatrix} \quad \& \quad B = \frac{\partial G}{\partial Q} = \begin{pmatrix} 0 & 0 & \beta \\ 0 & v & u \\ 1 & 0 & 2v \end{pmatrix}$$

معادلات ناویر-استوکس تراکم ناپذیر

• روش تراکم پذیری مصنوعی (Artificial Compressibility Method)

- روش ضمنی اویلر

$$\frac{\partial Q^{n+1}}{\partial t} + \frac{\partial F^{n+1}}{\partial x} + \frac{\partial G^{n+1}}{\partial y} = \frac{1}{\text{Re}} N \nabla^2 Q^{n+1}$$

$$\frac{\Delta Q^n}{\Delta t} + \frac{\partial (F^n + A^n \Delta Q^n)}{\partial x} + \frac{\partial (G^n + B^n \Delta Q^n)}{\partial y} = \frac{1}{\text{Re}} N \nabla^2 (Q^n + \Delta Q^n)$$

$$\left\{ I + \Delta t \left[\frac{\partial A^n}{\partial x} + \frac{\partial B^n}{\partial y} - \frac{1}{\text{Re}} N \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) \right] \right\} \Delta Q^n = RHS$$

$$RHS = \Delta t \left[-\frac{\partial F^n}{\partial x} - \frac{\partial G^n}{\partial y} + \frac{1}{\text{Re}} N \nabla^2 Q^n \right]$$

$$\left\{ I + \Delta t \left[\frac{\partial A^n}{\partial x} - \frac{1}{\text{Re}} N \frac{\partial^2}{\partial x^2} \right] \right\} \left\{ I + \Delta t \left[\frac{\partial B^n}{\partial y} - \frac{1}{\text{Re}} N \frac{\partial^2}{\partial y^2} \right] \right\} \Delta Q^n = RHS$$

معادلات ناویر-استوکس تراکم‌ناپذیر

• روش تراکم‌پذیری مصنوعی (Artificial Compressibility Method)

- روش ضمنی اوایلر

$$\left\{ I + \Delta t \left[\frac{\partial A^n}{\partial x} - \frac{1}{\text{Re}} N \frac{\partial^2}{\partial x^2} \right] \right\} \left\{ I + \Delta t \left[\frac{\partial B^n}{\partial y} - \frac{1}{\text{Re}} N \frac{\partial^2}{\partial y^2} \right] \right\} \Delta Q^n = RHS$$

- گسسته‌سازی تفاضل مرکزی (با اضافه کردن عبارات میرایی عددی صریح و ضمنی)

Step 1:

$$\left\{ I + \Delta t \left[\frac{\partial A^n}{\partial x} - \frac{1}{\text{Re}} N \frac{\partial^2}{\partial x^2} \right] - \varepsilon_i (\Delta x)^2 \frac{\partial^2}{\partial x^2} \right\} \Delta Q^* = RHS - \varepsilon_e \left[(\Delta x)^4 \frac{\partial^4}{\partial x^4} + (\Delta y)^4 \frac{\partial^4}{\partial y^4} \right] Q^n$$

Step 2:

$$\left\{ I + \Delta t \left[\frac{\partial B^n}{\partial y} - \frac{1}{\text{Re}} N \frac{\partial^2}{\partial y^2} \right] - \varepsilon_i (\Delta y)^2 \frac{\partial^2}{\partial y^2} \right\} \Delta Q^n = \Delta Q^*$$

Step 3:

$$Q^{n+1} = Q^n + \Delta Q^n$$

معادلات ناویر-استوکس تراکم ناپذیر

• روش تراکم پذیری مصنوعی (Artificial Compressibility Method)

Step 1 :

- شرایط مرزی

$$\bar{A}\Delta Q_{i-1,j}^* + \bar{B}\Delta Q_{i,j}^* + \bar{C}\Delta Q_{i+1,j}^* = RHS \quad \& \quad Q^* = \begin{pmatrix} p^* \\ u^* \\ v^* \end{pmatrix} \quad \& \quad \Delta Q^* = \begin{pmatrix} \Delta p^* \\ \Delta u^* \\ \Delta v^* \end{pmatrix}$$

Step 2 :

$$\bar{\bar{A}}\Delta Q_{i,j-1} + \bar{\bar{B}}\Delta Q_{i,j} + \bar{\bar{C}}\Delta Q_{i,j+1} = \Delta Q^* \quad \& \quad Q = \begin{pmatrix} p \\ u \\ v \end{pmatrix} \quad \& \quad \Delta Q = \begin{pmatrix} \Delta p \\ \Delta u \\ \Delta v \end{pmatrix}$$

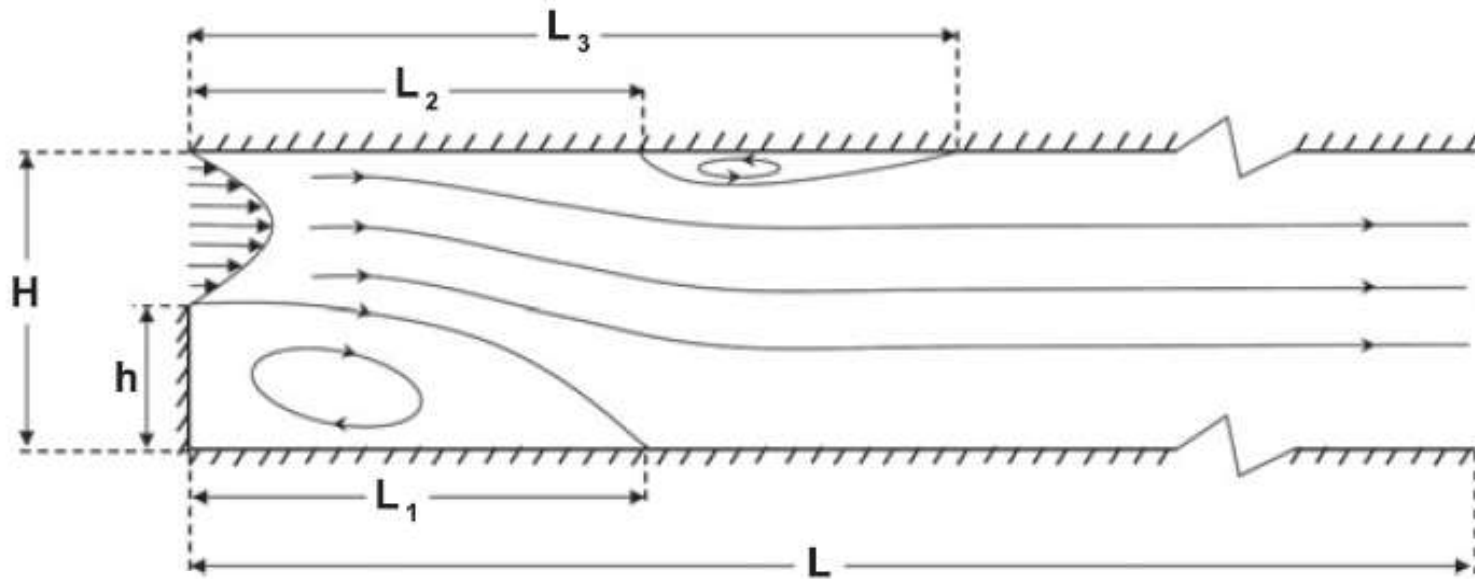
Step 3 :

$$Q^{n+1} = Q^n + \Delta Q^n$$

معادلات ناویر-استوکس تراکم‌ناپذیر

- روش تراکم‌پذیری مصنوعی (Artificial Compressibility Method)

- شرایط مرزی برای مساله پله رو به پشت



Geometry of backward-facing step including defined important flow parameters.

معادلات ناویر-استوکس تراکم ناپذیر

• روش تراکم پذیری مصنوعی (Artificial Compressibility Method)

Step 1 :

- شرایط مرزی

$$\bar{A} \Delta Q_{i-1,j}^* + \bar{B} \Delta Q_{i,j}^* + \bar{C} \Delta Q_{i+1,j}^* = RHS \quad \& \quad Q^* = \begin{pmatrix} p^* \\ u^* \\ v^* \end{pmatrix} \quad \& \quad \Delta Q^* = \begin{pmatrix} \Delta p^* \\ \Delta u^* \\ \Delta v^* \end{pmatrix}$$

At wall: $\begin{cases} u^* = v^* = 0 \\ \frac{\partial p^*}{\partial n} = 0 \end{cases} \rightarrow \Delta Q_{1,j}^* = \begin{pmatrix} \Delta p_{1,j}^* \\ \Delta u_{1,j}^* \\ \Delta v_{1,j}^* \end{pmatrix} = \begin{pmatrix} \Delta p_{1,j}^* \\ 0 \\ 0 \end{pmatrix} \quad \& \quad \Delta p_{1,j}^* = \frac{1}{3} (4\Delta p_{2,j}^* - \Delta p_{3,j}^*)$

Step 2 :

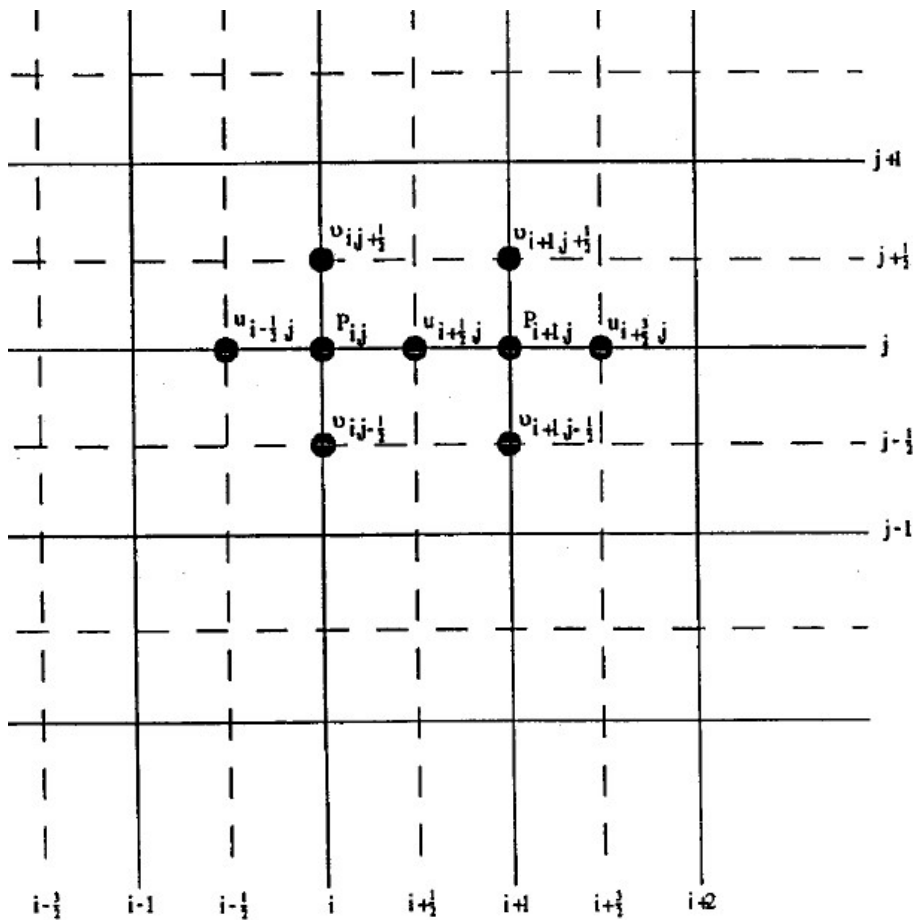
$$\bar{\bar{A}} \Delta Q_{i,j-1} + \bar{\bar{B}} \Delta Q_{i,j} + \bar{\bar{C}} \Delta Q_{i,j+1} = \Delta Q^*$$

At wall: $\begin{cases} u = v = 0 \\ \frac{\partial p}{\partial n} = 0 \end{cases} \rightarrow \Delta Q_{i,1} = \begin{pmatrix} \Delta p_{i,1} \\ \Delta u_{i,1} \\ \Delta v_{i,1} \end{pmatrix} = \begin{pmatrix} \Delta p_{i,1} \\ 0 \\ 0 \end{pmatrix} \quad \& \quad \Delta p_{i,1} = \frac{1}{3} (4\Delta p_{i,2} - \Delta p_{i,3})$

معادلات ناویر-استوکس تراکم ناپذیر

• شبکه جابجاشده (Staggered Grid)

- استفاده از دو شبکه (شبکه اولیه/اصلی و شبکه ثانویه)
- تعریف فشار روی شبکه اولیه و تعریف مولفه‌های سرعت روی وجوه سلول‌های شبکه ثانویه
- کوپل شدن متغیرها و افزایش پایداری



معادلات ناویر-استوکس تراکم ناپذیر

• روش‌های اصلاح فشار (Pressure Correction Methods)

Marker-and-Cell Method (Harlow and Welch (1965) & Welch et al (1966)

Solving the time dependent free surface flow by tracking the paths of the fictitious massless marker particles introduced on the free surface

Two steps:

1) Solving the momentum eqs. for (u,v) using the best current estimate of the pressure distribution

$$\frac{\partial u}{\partial t} + \frac{\partial(u^2 + p)}{\partial x} + \frac{\partial(uv)}{\partial y} = \frac{1}{\text{Re}} \nabla^2 u$$

$$\frac{\partial v}{\partial t} + \frac{\partial(uv)}{\partial x} + \frac{\partial(v^2 + p)}{\partial y} = \frac{1}{\text{Re}} \nabla^2 v$$

معادلات ناویر-استوکس تراکم ناپذیر

• روش‌های اصلاح فشار (Pressure Correction Methods)

Marker-and-Cell Method (Harlow and Welch (1965) & Welch et al (1966)

2) Improving the pressure distribution using the Poisson equation

$$\frac{\partial}{\partial y}(x - mom.) + \frac{\partial}{\partial x}(y - mom.) \rightarrow \nabla^2 p = \frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 p}{\partial y^2} = S_p - \frac{\partial D}{\partial t}$$

$$S_p = -\frac{\partial^2}{\partial x^2}(u^2) - \frac{\partial^2}{\partial y^2}(v^2) - 2\frac{\partial^2}{\partial x \partial y}(uv) + \frac{1}{Re}\nabla^2 D$$

$$D = \vec{\nabla} \cdot \vec{V} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \quad \& \quad \text{for incompressible flow: } D = \vec{\nabla} \cdot \vec{V} = 0$$

In the MAC method, D is evaluated within (*) to prevent the error accumulation in the process of the iterative solution of the equation, and prevent the nonlinearity instability.

The MAC method does not require specification of any B.C for the pressure on a staggered grid.

معادلات ناویر-استوکس تراکم ناپذیر

• روش‌های اصلاح فشار (Pressure Correction Methods)

Markel-and-Cell Method (Harlow and Welch (1965) & Wlch et al (1966)

Numerical Procedure

1) x-mom. is applied on the secondary grid at the point $(i+1/2, j)$

$$\frac{u_{i+\frac{1}{2},j}^{n+1} - u_{i+\frac{1}{2},j}^n}{\Delta t} + \frac{(u^2)_{i+1,j}^n - (u^2)_{i,j}^n}{\Delta x} + \frac{(uv)_{i+\frac{1}{2},j+\frac{1}{2}}^n - (uv)_{i+\frac{1}{2},j-\frac{1}{2}}^n}{\Delta y} + \frac{p_{i+1,j}^n - p_{i,j}^n}{\Delta x} = \frac{1}{\text{Re}} \left(\frac{u_{i-\frac{1}{2},j}^{n+1} - 2u_{i+\frac{1}{2},j}^n + u_{i+\frac{3}{2},j}^n}{(\Delta x)^2} + \frac{u_{i+\frac{1}{2},j-1}^{n+1} - 2u_{i+\frac{1}{2},j}^n + u_{i+\frac{1}{2},j+1}^n}{(\Delta y)^2} \right)$$

2) y-mom. is applied at the grid point $(i, j+1/2)$

$$\frac{v_{i,j+\frac{1}{2}}^{n+1} - v_{i,j+\frac{1}{2}}^n}{\Delta t} + \frac{(uv)_{i+\frac{1}{2},j+\frac{1}{2}}^n - (uv)_{i-\frac{1}{2},j-\frac{1}{2}}^n}{\Delta y} + \frac{(v^2)_{i,j+1}^n - (v^2)_{i,j}^n}{\Delta y} + \frac{p_{i,j+1}^n - p_{i,j}^n}{\Delta y} = \frac{1}{\text{Re}} \left(\frac{v_{i-1,j+\frac{1}{2}}^{n+1} - 2v_{i,j+\frac{1}{2}}^n + v_{i+1,j+\frac{1}{2}}^n}{(\Delta x)^2} + \frac{v_{i,j-\frac{1}{2}}^{n+1} - 2v_{i,j+\frac{1}{2}}^n + v_{i,j+\frac{3}{2}}^n}{(\Delta y)^2} \right)$$

All the values appearing in terms of the primary grid point must be replaced by their approximate values on the secondary grid

$$u_{i+1,j} = \frac{1}{2} \left(u_{i+\frac{1}{2},j} + u_{i+\frac{3}{2},j} \right) \quad \& \quad u_{i,j} = \frac{1}{2} \left(u_{i+\frac{1}{2},j} + u_{i-\frac{1}{2},j} \right) \quad \& \quad (uv)_{i+\frac{1}{2},j+\frac{1}{2}} = \frac{1}{4} \left(u_{i+\frac{1}{2},j} + u_{i+\frac{1}{2},j+1} \right) \left(v_{i,j+\frac{1}{2}} + v_{i+1,j+\frac{1}{2}} \right)$$

معادلات ناویر-استوکس تراکم ناپذیر

• روش‌های اصلاح فشار (Pressure Correction Methods)

Markel-and-Cell Method (Harlow and Welch (1965) & Welch et al (1966)

Numerical Procedure

3) Use of the Poisson equation for the pressure

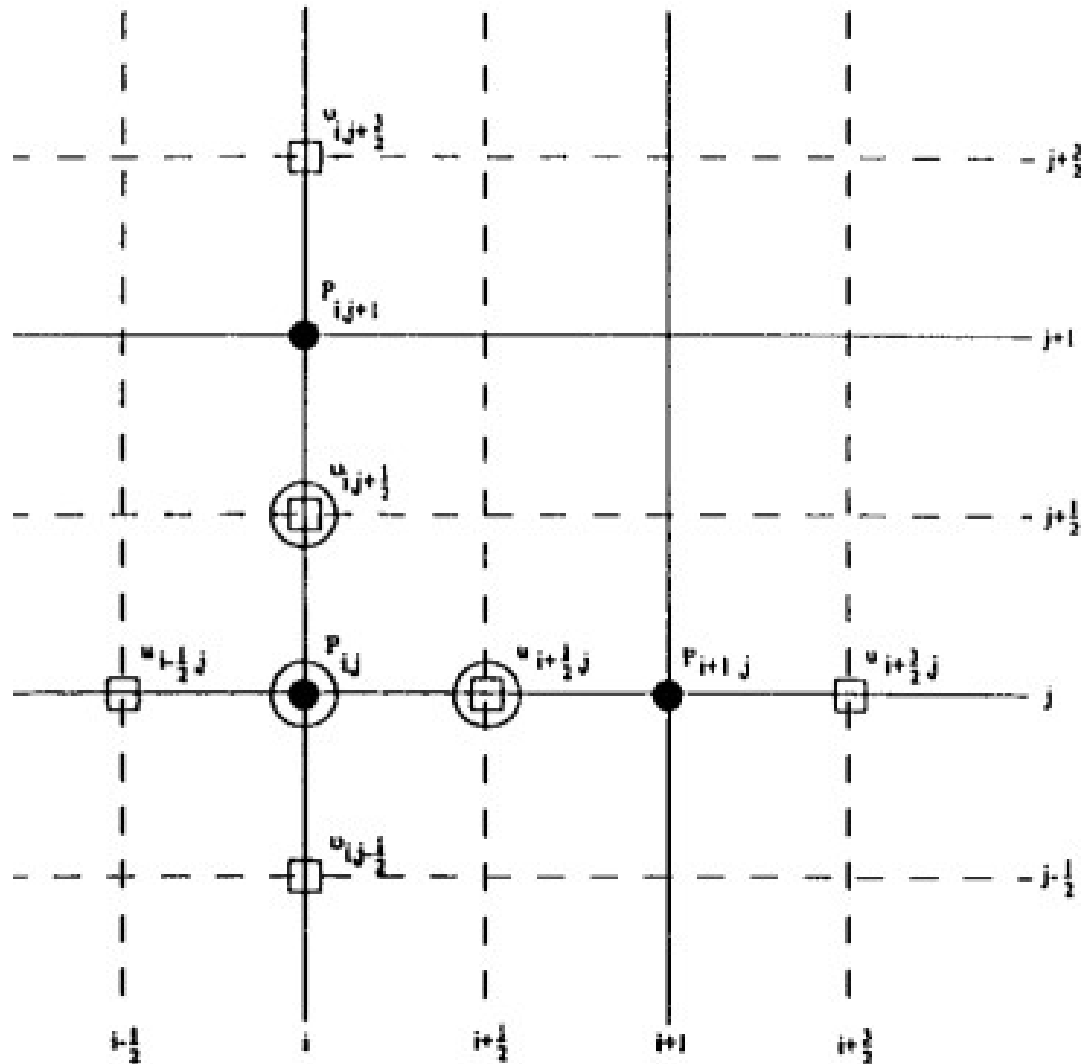
$$p_{i,j}^{k+1} = \frac{1}{2(1 + \beta^2)} \left[p_{i+1,j}^k + p_{i-1,j}^{k+1} + \beta^2 (p_{i,j+1}^k + p_{i,j-1}^{k+1}) \right] + RHS_{i,j} \quad \text{Point Gauss-Seidel Scheme}$$

$$RHS_{i,j} = \frac{D_{i,j}^{n+1} - D_{i,j}^n}{\Delta t} - \frac{(u^2)_{i-1,j}^n - 2(u^2)_{i,j}^n + (u^2)_{i+1,j}^n}{(\Delta x)^2} - \frac{(v^2)_{i,j-1}^n - 2(v^2)_{i,j}^n + (v^2)_{i,j+1}^n}{(\Delta y)^2} \\ - 2 \frac{(uv)_{i+\frac{1}{2},j+\frac{1}{2}}^n - (uv)_{i-\frac{1}{2},j+\frac{1}{2}}^n - (uv)_{i+\frac{1}{2},j-\frac{1}{2}}^n + (uv)_{i-\frac{1}{2},j-\frac{1}{2}}^n}{(\Delta x \Delta y)} + \frac{1}{\text{Re}} \left(\frac{D_{i-1,j}^n - 2D_{i,j}^n + D_{i+1,j}^n}{(\Delta x)^2} + \frac{D_{i,j-1}^n - 2D_{i,j}^n + D_{i,j+1}^n}{(\Delta y)^2} \right)$$

Note: $D_{i,j}^{n+1}$ is set to zero; the continuity requirement at $n+1$ time level is enforced.

معادلات ناویر-استوکس تراکم ناپذیر

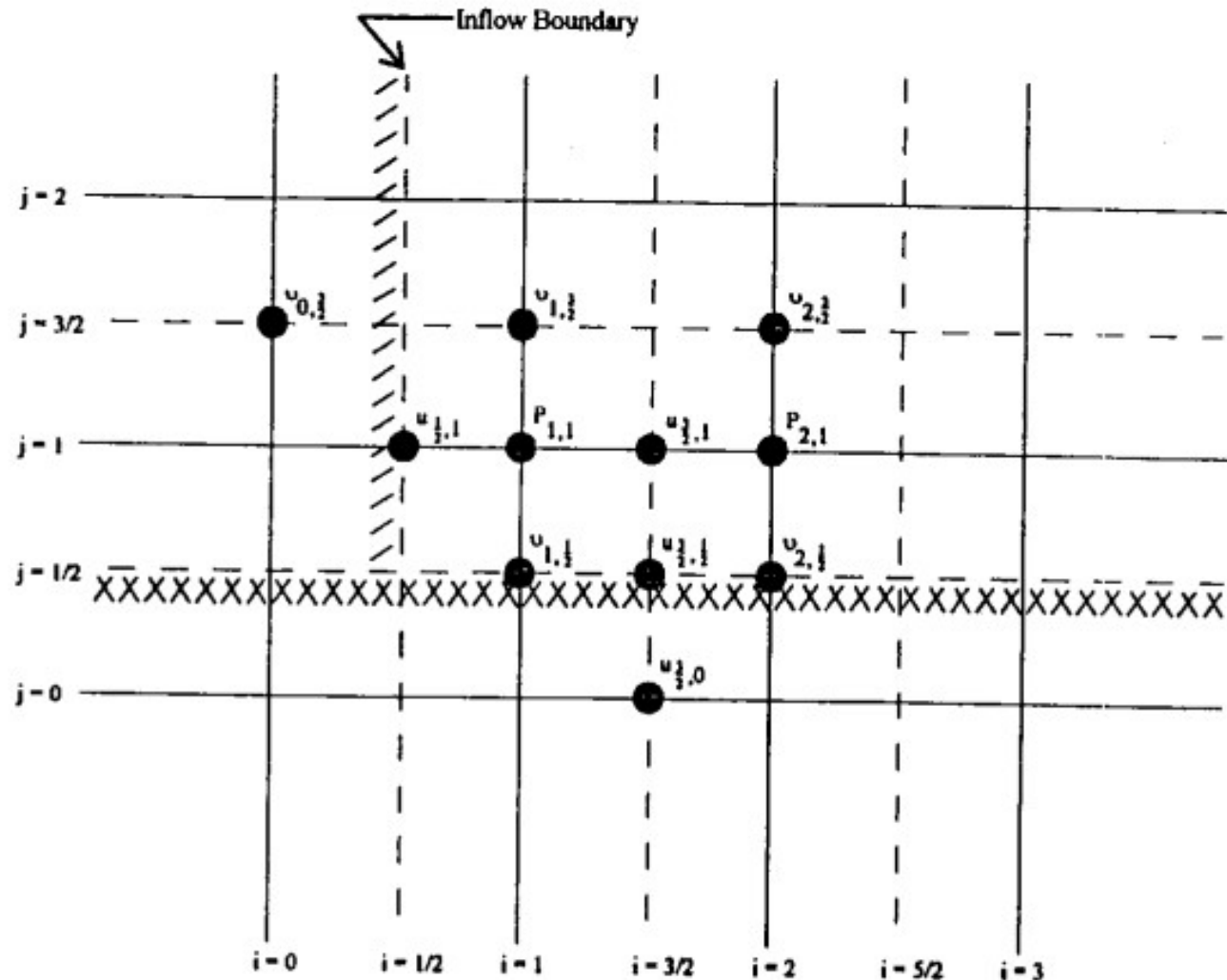
Assignment of the variables on the staggered grid.



معادلات ناویر-استوکس تراکم ناپذیر

Implementation of B.Cs

Only specification of velocity boundary conditions is required



معادلات ناویر-استوکس تراکم ناپذیر

Implementation of B.Cs

No-slip BCs:

$$v_{1,\frac{1}{2}} = v_{2,\frac{1}{2}} = \dots = 0 \quad \& \quad u_{\frac{3}{2},\frac{1}{2}} = 0$$
$$u_{\frac{3}{2},\frac{1}{2}} = \frac{1}{2} \left(u_{\frac{3}{2},0} + u_{\frac{3}{2},1} \right) = 0 \quad \rightarrow \quad u_{\frac{3}{2},0} = -u_{\frac{3}{2},1}$$

Inflow:

u components are specified, $u_{\frac{1}{2},1}, u_{\frac{1}{2},2}, \dots$

v components are approximated by extrapolation

$$v_{0,\frac{3}{2}} = 2v_{\frac{1}{2},\frac{3}{2}} - v_{1,\frac{3}{2}} \quad \& \quad v_{\frac{1}{2},\frac{3}{2}} = \text{specified inflow boundary}$$

معادلات ناویر-استوکس تراکم ناپذیر

• روش‌های اصلاح فشار (Pressure Correction Methods)

Projection or Fractional Step Method (Chorin (1968) & Temam (1969))

- Regular grid (non-staggered grid)

Two steps:

1) Omitting the pressure gradient terms from the momentum Eqs. and advancing in time to obtain $\vec{V}^*$

$$\frac{\partial \vec{V}^*}{\partial t} + (\vec{V}^* \cdot \vec{\nabla}) \vec{V}^* = \frac{1}{\text{Re}} \nabla^2 \vec{V}^* \quad (\text{a})$$

2) Correcting the velocity field by accounting for the pressure gradient and the continuity eq.

$$\frac{\vec{V}^{n+1} - \vec{V}^*}{\Delta t} + \vec{\nabla} p^{n+1} = 0 \quad (\text{b})$$

$$\vec{\nabla} \cdot \left(\frac{\vec{V}^{n+1} - \vec{V}^*}{\Delta t} + \vec{\nabla} p^{n+1} \right) = 0 \quad \rightarrow \quad \frac{\vec{\nabla} \cdot \vec{V}^{n+1} - \vec{\nabla} \cdot \vec{V}^*}{\Delta t} + \vec{\nabla} \cdot \vec{\nabla} p^{n+1} = 0$$

$$\vec{\nabla} \cdot \vec{V}^{n+1} = 0 \quad \rightarrow \quad \nabla^2 p^{n+1} = \frac{\vec{\nabla} \cdot \vec{V}^*}{\Delta t} \quad (\text{c})$$

معادلات ناویر-استوکس تراکم‌ناپذیر

• روش‌های اصلاح فشار (Pressure Correction Methods)

Projection or Fractional Step Method (Chorin (1968) & Temam (1969))

Procedure:

- 1) Solving the momentum equations without the pressure gradient terms to calculate V^* (a)
- 2) Solving the Poisson equation for the pressure (c)
- 3) Evaluating the velocity field (b)

معادلات ناویر-استوکس تراکم ناپذیر

• روش‌های اصلاح فشار (Pressure Correction Methods)

SIMPLE (Semi-Implicit Method for Pressure Linked Equation) family of methods

Caretto et al. (1972) & Patankar and Spalding (1972)

- A cyclic series of guess-and-correct operations

Procedure:

1) The velocity field are first calculated from the momentum equations using a guessed pressure field

2) The pressure and velocity are corrected to satisfy the continuity eq.

$$p = p_0 + p' \quad \& \quad u = u_0 + u' \quad \& \quad v = v_0 + v'$$

p = actual pressure & p_0 = estimated (intermediate) pressure & p' = pressure correction

Using approximate forms of the momentum eqs. and assuming the velocity corrections to be zero at the previous iteration step:

$$\begin{cases} \rho \frac{\partial u'}{\partial t} = -\frac{\partial p'}{\partial x} \\ \rho \frac{\partial v'}{\partial t} = -\frac{\partial p'}{\partial y} \end{cases} \rightarrow \begin{cases} u' = -A \frac{\partial p'}{\partial x} \\ v' = -A \frac{\partial p'}{\partial y} \end{cases} \quad \& \quad A = \frac{\Delta t}{\rho} \quad \& \quad \Delta t = \text{the fictitious time increment}$$

معادلات ناویر-استوکس تراکم ناپذیر

• روش های اصلاح فشار (Pressure Correction Methods)

SIMPLE (Semi-Implicit Method for Pressure Linked Equation) family of methods

Using the continuity equation:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \quad \text{enforce the mass conservation}$$

$$u = u_0 + u' \quad \& \quad v = v_0 + v' \quad \& \quad \vec{V}_0 = (u_0, v_0)$$

$$\frac{\partial u_0}{\partial x} + \frac{\partial v_0}{\partial y} + \frac{\partial u'}{\partial x} + \frac{\partial v'}{\partial y} = 0$$

$$\vec{\nabla} \cdot \vec{V}_0 + \left(-A \frac{\partial^2 p'}{\partial x^2} - A \frac{\partial^2 p'}{\partial y^2} \right) = 0$$

$$\nabla^2 p' = \frac{1}{A} \vec{\nabla} \cdot \vec{V}_0 \quad (*)$$

If the estimated velocity vector satisfies the continuity at every point, then the pressure correction is zero at every point

معادلات ناویر-استوکس تراکم ناپذیر

• روش های اصلاح فشار (Pressure Correction Methods)

SIMPLE (Semi-Implicit Method for Pressure Linked Equation) family of methods

Procedure:

- 1) Guess the pressure p_0 at each grid point
- 2) Solve the momentum equations to find the velocity (u_0, v_0)
- 3) Solve the pressure-correction equation (*) to find p' at each grid point
- 4) Correct the pressure and velocity

$$p = p_0 + p'$$

$$\begin{cases} \rho \frac{\partial u'}{\partial t} = -\frac{\partial p'}{\partial x} \\ \rho \frac{\partial v'}{\partial t} = -\frac{\partial p'}{\partial y} \end{cases} \rightarrow \begin{cases} u = u_0 - \frac{A}{2\Delta x} (p'_{i+1,j} - p'_{i-1,j}) \\ v = v_0 - \frac{A}{2\Delta y} (p'_{i,j+1} - p'_{i,j-1}) \end{cases}$$

- 5) Replacing the previous values for (p_0, u_0, v_0) with the new corrected values (p, u, v) and return to step 2)

معادلات ناویر-استوکس تراکم‌ناپذیر

• روش‌های اصلاح فشار (Pressure Correction Methods)

SIMPLE (Semi-Implicit Method for Pressure Linked Equation) family of methods

Problems:

- 1) Overestimating the value of p' due to an approximate Poisson equation ($\nabla^2 p' = \dots$)
- 2) Low convergence

Under-relaxation for p'

$$p = p_0 + \alpha_p p' \quad \& \quad u = u_0 + \alpha_u u' \quad \& \quad v = v_0 + \alpha_v v'$$

α = an under-relaxation constant ≈ 0.8

معادلات ناویر-استوکس تراکم ناپذیر

• روش های اصلاح فشار (Pressure Correction Methods)

SIMPLER (SIMPLE Revised)

Determination of the optimum α_p

Procedure:

- 1) The velocity corrections are computed in the same manner as in SIMPLE
- 2) A complete Poisson equation ($\nabla^2 p = \dots$) for the pressure is used
 - The velocity field is guessed initially instead of the pressure field (estimating a reasonable velocity field)
 - The pressures computed in SIMPLER are closed to the correct pressures and thus the need for under-relaxation is greatly reduced

موفق باشید.